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UNIT → 4

$\underbrace{\text{CONTINUOUS}}_{}$ RANDOM VARIABLE

exact value ni pata, has interval pta hai

→ Probablity Density Function (PDF)
Let X be a continuous random variable with distribution function $F(x) = P(-\infty < X \leq x)$. Then a function $f(x)$ is the PDF of X if $f(x)$ is integrable on interval $[a, x]$ $\forall$ a and if

$$F(x) = \int_{-\infty}^{x} f(t)\,dt$$

★
→ Properties of PDF
- $f(x) \geq 0$
- $\int_{-\infty}^{\infty} f(x)\,dx = 1$

Q A lifetime of an bulb is a random variable with PDF $f(x) = Kx^2$ where x is in years min & max lifetimes are 1 & 2 years resp.
$K = ?$

$$\Rightarrow \int_{1}^{2} Kx^2\,dx = 1$$

min se max value tak PDF Ka $\int$ always $= 1$

$$\left.\frac{Kx^3}{3}\right|_{1}^{2} = 1$$

$$\frac{8K}{3} - \frac{K}{3} = 1$$

$$K = \frac{3}{7}$$

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→ Properties of distribution function

- $F(a) - \lim_{x \to a^-} F(x) = P(X = a)$

- For real nos a & b $(a < b)$

$$P(a \le X < b) = P(a < x < b) = P(a < X \le b) = \int_a^b f(t)\,dt$$
$$= F(b) - F(a)$$

- $f(x)$ is continous and $f(x) = F'(x)$

→ Expectation (Mean) of continuous random variable

$$E(x) = \int_{-\infty}^{\infty} x f(x)\,dx$$

- $E(a) = a$ where $a \to$ constant
- $E(aX) = a E(X)$
- $E(X \pm Y) = E(X) \pm E(Y)$
- $E(XY) = E(X) \cdot E(Y)$ if X & Y are independent

→ Variance & standard deviation

$$Var(X) = [E(X - E(X))]^2$$

$$\sigma = E(X - E(X))$$

- $Var(X) = E(X^2) - [E(X)]^2$
- $Var(aX + b) = a^2 Var(X)$
- $Var(K) = 0$ where $K \to$ constant

Q $f(x) = K(x-1)(2-x)$ for $1 \le x \le 2$ Find K, $F(x)$, $P\left(\frac{5}{4} \le x \le \frac{3}{2}\right) = ?$

$$\int_1^2 K(-x^2 + 3x - 2)\,dx = 1$$

$$-K\left[\frac{x^3}{3} - \frac{3x^2}{2} + 2x\right]_1^2 = 1$$

$$K\left[\frac{7}{3}-\frac{9}{2}+2\right] = -1$$

$$K\left[\frac{14-27+12}{6}\right] = -1$$

$$K = 6$$

$$f(x) = \begin{cases} 6\,(x-1)(2-x) & 1 \le x \le 2 \\ 0 & \text{otherwise} \end{cases}$$

$$F(x) = \int_{-\infty}^{x} f(x)\,dx$$

at $x < 1$, $f(x) = 0$

$$= \int_{-\infty}^{x} 0\,dx = 0$$

at $1 \le x \le 2$

$$F(x) = \int_{-\infty}^{1} f(x)\,dx + \int_{1}^{x} f(x)\,dx$$

$$= 0 + \int_{1}^{x} f(x)\,dx$$

$$= \int_{1}^{x} 6\,(x-1)(2-x)\,dx$$

$$= 5 - 12x + 9x^2 - 2x^3$$

at $x > 2$

$$F(x) = \int_{-\infty}^{1} f(x)\,dx + \int_{1}^{2} f(x)\,dx + \int_{2}^{x} f(x)\,dx$$

$$= 0 + \int_{1}^{2} f(x)\,dx + 0$$

$$= 1$$

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$$\Rightarrow F(x) = \begin{cases} 0 & x < 1 \\ 5 - 12x + 9x^2 - 2x^3 & 1 \le x \le 2 \\ 1 & x > 2 \end{cases}$$

NOW,

$$P\left(\frac{5}{4} \le X \le \frac{3}{2}\right)$$

$$\int_{5/4}^{3/2} 6(x-1)(2-x)\,dx \quad = F\left(\frac{3}{2}\right) - F\left(\frac{5}{4}\right)$$

either do this or this

$\frac{3}{2} = 1.5$ & $\frac{5}{4} = 1.25$

$$F\left(\frac{3}{2}\right) = 5 - 12\left(\frac{3}{2}\right) + 9\left(\frac{3}{2}\right)^2 - 2\left(\frac{3}{2}\right)^3 =$$

$$F\left(\frac{5}{4}\right) = 5 - 12\left(\frac{5}{4}\right) + 9\left(\frac{5}{4}\right)^2 - 2\left(\frac{5}{4}\right)^3 =$$

$$\text{Ans} = \frac{11}{32}$$

PYQ

Q A cont. Random variable X have values between 0 & 4 has density func $P(X) = \frac{1}{2} - ax$

1. Find a
2. Find $P(1 < x < 2)$

$$\int_0^4 P(x)\,dx = 1$$

$$\Rightarrow \int_0^4 \left(\frac{1}{2} - ax\right) dx = 1$$

$$\Rightarrow \left.\frac{x}{2} - \frac{ax^2}{2}\right|_0^4 = 1$$

$$\Rightarrow 2 - 8a = 1$$

$$8a = 1 \quad \Rightarrow \quad a = 1/8$$

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Hence, $P(x) = \begin{cases} \frac{1}{2} - \frac{x}{8} & 0 \le x \le 4 \\ 0 & \text{otherwise} \end{cases}$

For $x < 0$

$F(x) = \int_{-\infty}^{0} P(x)\,dx = 0$

For $0 \le x \le 1$

$F(x) = \int_{-\infty}^{0} P(x)\,dx + \int_{1}^{x} \frac{1}{2} - \frac{x}{8}\,dx$

$= \frac{x}{2} - \frac{x^2}{16} \Big|_{1}^{x}$

$= \frac{x}{2} - \frac{x^2}{16} - \frac{1}{2} + \frac{1}{16}$

$= \frac{x}{2} - \frac{x^2}{16} - \frac{7}{16}$

For $1 \le x \le 2$

$F(x) = \int_{-\infty}^{0} P(x)\,dx + \int_{1}^{2} P(x)\,dx + \int_{2}^{x} P(x)\,dx$

$= \int_{1}^{2} \frac{1}{2} - \frac{x}{8} + \int_{2}^{x} \frac{1}{2} - \frac{x}{8}$

$= \frac{x}{2} - \frac{x^2}{16} \Big|_{1}^{2} + \frac{x}{2} - \frac{x^2}{16} \Big|_{2}^{x}$

$= \frac{1}{2} - \frac{3}{16} + \frac{x}{2} - \frac{x^2}{16} - 1 + \frac{1}{4}$

$= \frac{x}{2} - \frac{x^2}{16} - \frac{7}{16}$

$F(1) = 1/2 - 1/16 - 7/16 = 0$

$F(2) = 1 - 1/4 - 7/16 = 5/16$

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$$P(1 \le X \le 2) = \int_1^2 P(x)\,dx = F(2) - F(1) = \frac{5}{16}$$

Q $f(x) = \begin{cases} x & 0 \le x < 1 \\ K - x & 1 \le x \le 2 \\ 0 & \text{otherwise} \end{cases}$

Find K, $F(x)$ & $P\left(\frac{1}{2} \le x \le \frac{3}{2}\right)$

$$\int_0^1 x + \int_1^2 K - x = 1$$

$$\frac{1}{2} + Kx - \frac{x^2}{2}\Big|_1^2 = 1$$

$$K - \frac{3}{2} = \frac{1}{2} \qquad K = \frac{3}{2} + \frac{1}{2} = 2$$

$f(x) = \begin{cases} x & 0 \le x < 1 \\ 2 - x & 1 \le x \le 2 \\ 0 & \text{otherwise} \end{cases}$

For $x < 0$

$$F(x) = \int_{-\infty}^{0} 0\,dx = 0$$

For $0 \le x < 1$

$$F(x) = \int_{-\infty}^{0} 0\,dx + \int_0^x x\,dx = \frac{x^2}{2}$$

For $1 \le x \le 2$

$$F(x) = \int_{-\infty}^{0} 0\,dx + \int_0^1 x\,dx + \int_1^x 2 - x$$

$$= \frac{1}{2} + 2x - 2 - \frac{x^2}{2} + \frac{1}{2} = 2x - \frac{x^2}{2} - 1$$

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For $x > 2$

$$F(x) = \int_{-\infty}^{0} 0\,dx + \int_{0}^{1} x\,dx + \int_{1}^{2} 2-x\,dx + \int_{2}^{x} 0\,dx$$

$$= \frac{1}{2} + \left[2x - \frac{x^2}{2}\right]_1^2 = \frac{1}{2} + 4 - 2 - 2 + \frac{1}{2}$$

$$= 1$$

$$F(x) = \begin{cases} 0 & x < 0 \\ x^2/2 & 0 \le x \le 1 \\ 2x - x^2/2 - 1 & 1 \le x \le 2 \\ 1 & x > 2 \end{cases}$$

Now,

$$P\left(\frac{1}{2} \le x \le \frac{3}{2}\right) = F\left(\frac{3}{2}\right) - F\left(\frac{1}{2}\right)$$

$\frac{1}{2} = 0.5$ $\qquad$ $\frac{3}{2} = 1.5$

$F\left(\frac{1}{2}\right) = \frac{1}{8}$ $\qquad$ $F\left(\frac{3}{2}\right) = 3 - \frac{9}{8} - 1 = \frac{7}{8}$

$$P_{req} = \frac{7}{8} - \frac{1}{8} = \frac{6}{8} = \frac{3}{4}$$

→ **Moment generating function**

r^{th} moment of random variable X is given by

$$M_r = \int_{-\infty}^{\infty} e^{rx} \underbrace{f(x)}_{\text{PDF}}\,dx$$

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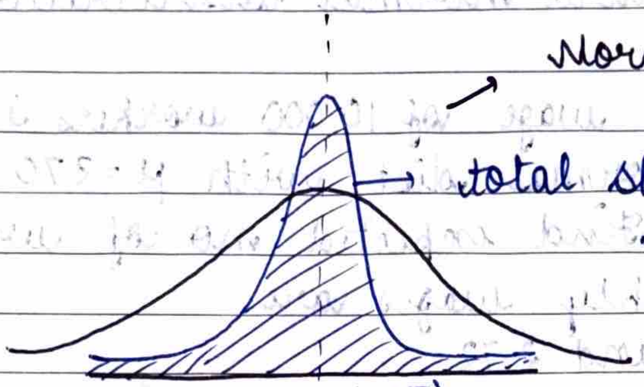
→ Normal Distribution (Gausian distribution)
For a distribution, tendency to accumulate about it's avg value



width depends on σ & height depends on μ

A random variable X is said to have normal distribution if it's PDF is given by

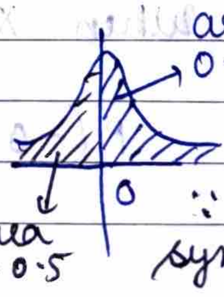
$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \, e^{-\frac{(x-\mu)^2}{2\sigma^2}} \qquad -\infty < x < \infty$$

$$x \sim N(\mu, \sigma)$$

- $f(x) \geq 0 \;\; \forall x$
  and
- $\int_{-\infty}^{\infty} f(x) = 1$

→ Standard Normal Distribution $X \sim N(0,1)$
Normal dist with $\mu = 0$ and $\sigma = 1$

$$f(x) = \frac{1}{\sqrt{2\pi}} \, e^{-\frac{x^2}{2}}$$



→ Theorem

If $X \sim N(\mu, \sigma)$ and $Z = \frac{x-\mu}{\sigma}$ then Z has standard normal distribution

Q If weekly wage of 10,000 workers in factory follows normal dist with $\mu = ₹70$ and $\sigma = ₹5$. Find expected no of workers whose weekly wages are

1) b/t ₹66 and ₹72
11) less than ₹66
111) more than ₹72

$$\left[\frac{1}{\sqrt{2\pi}}\int_0^z e^{-t^2/2}\,dt = 0.1554 \text{ when } z = 0.4\right.$$
$$\left.\text{and } 0.2881 \text{ when } z = 0.8\right]$$

Let $X \rightarrow$ random variable showing weekly wage for 1 worker

$$x \sim N(70, 5)$$

Now, $Z = \frac{x-\mu}{\sigma} = \frac{x-70}{5}$

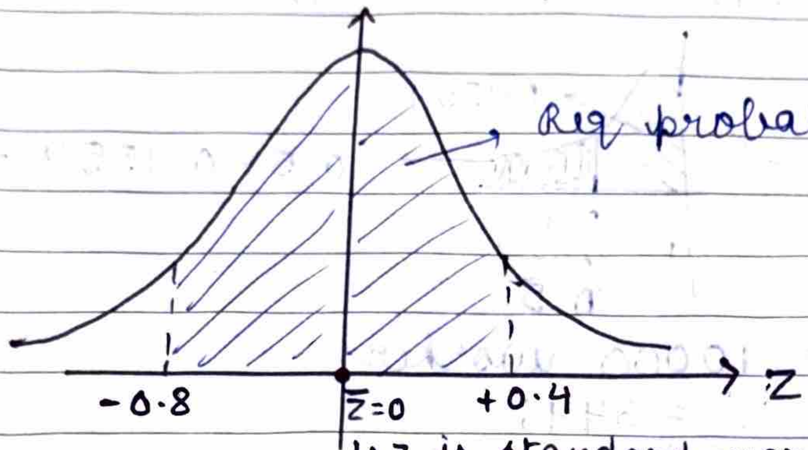
$\downarrow$ st. normal variable

1) $66 < X < 72$

when $X = 66$, $Z = \frac{66-70}{5} = -0.8$

when $X = 72$, $Z = \frac{2}{5} = 0.4$

$P(-0.8 < Z < 0.4)$



↳ z is standard normal variable

$$P(-0.8 < Z < 0.4) = \underbrace{P(-0.8 < Z < 0)}_{= P(0 < Z < 0.8)} + P(0 < Z < 0.4)$$

∵ curve is symmetrical

$$= P(0 < Z < 0.8) + P(0 < Z < 0.4)$$

$$= \int_0^{0.8} f(z)\,dz + \int_0^{0.4} f(z)\cdot dz$$

$$= \frac{1}{\sqrt{2\pi}} \int_0^{0.8} e^{-z^2/2}\,dz + \frac{1}{\sqrt{2\pi}} \int_0^{0.4} e^{-z^2/2}\,dz$$

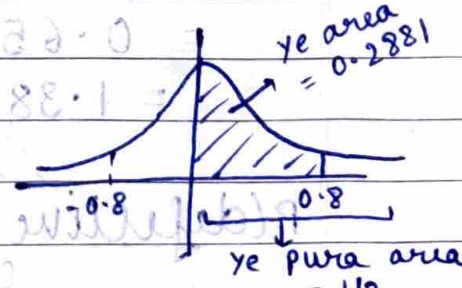
$$= 0.1544 + 0.2881$$

$$= 0.4425$$

For 10000, req wage = ₹ 4435

ii) $P(x < ₹65) = P(z < -0.8) = P(z > 0.8)$

$$= \int_{0.8}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-z^2/2}\,dz$$



$$= 0.5 - 0.2881 = 0.2119$$

For 10,000 workers

no = 2119

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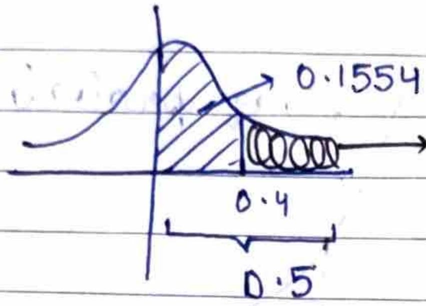
iii) $P(X > 72) = P(Z > 0.4)$



$0.5 - 0.1554 = 0.3446$

For 10000 workers
$= 3446$

Q Length of bolts produced by a machine is normally distributed with $\mu = 4$ & $\sigma = 0.5$ A bolt is defective if it's length doesn't lie in interval $(3.8, 4.3)$. Find % of defective bolts produced by machine.

$$\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{0.6} e^{-t^2/2}\,dt = 0.7257 \quad \& \quad \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{0.4} e^{-t^2/2}\,dt = 0.6554$$

$X \sim N(4, 0.5)$

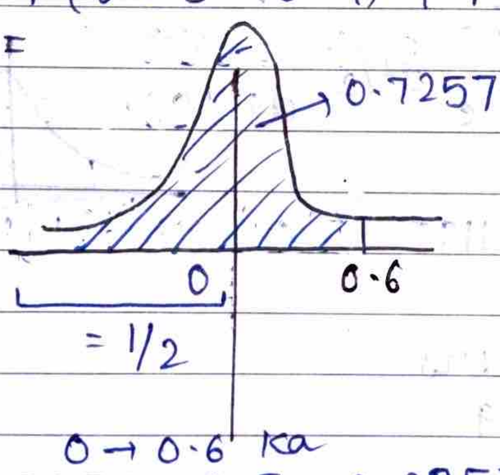
$$Z = \frac{x-\mu}{\sigma} = 2(x-4)$$

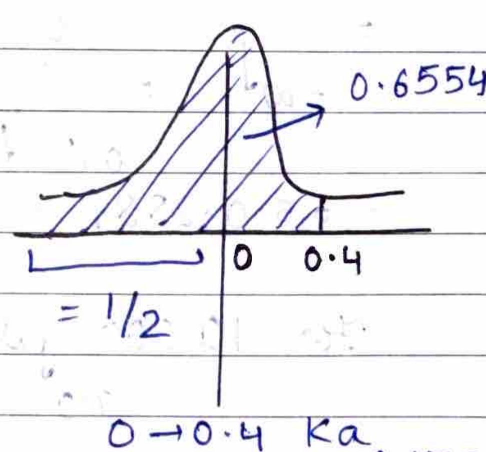
$P(\text{non defective bolt}) = P(3.8 < X < 4.3)$

$= P(-0.4 < Z < 0.6)$

$= P(-0.4 < Z < 0) + P(0 < x < 0.6)$

$= P(0 < Z < 0.4) + P(0 < x < 0.6)$





$0 \to 0.6$ ka
$0.7257 - 0.5 = 0.2257$

$0 \to 0.4$ ka
$0.6554 - 0.5 = 0.1554$

$P(\text{non defective}) = 0.2257 + 0.1554$
$= 0.3811$

$P(\text{defective}) = 1 - 0.3811 = 0.6189$
$= 61.89\%$

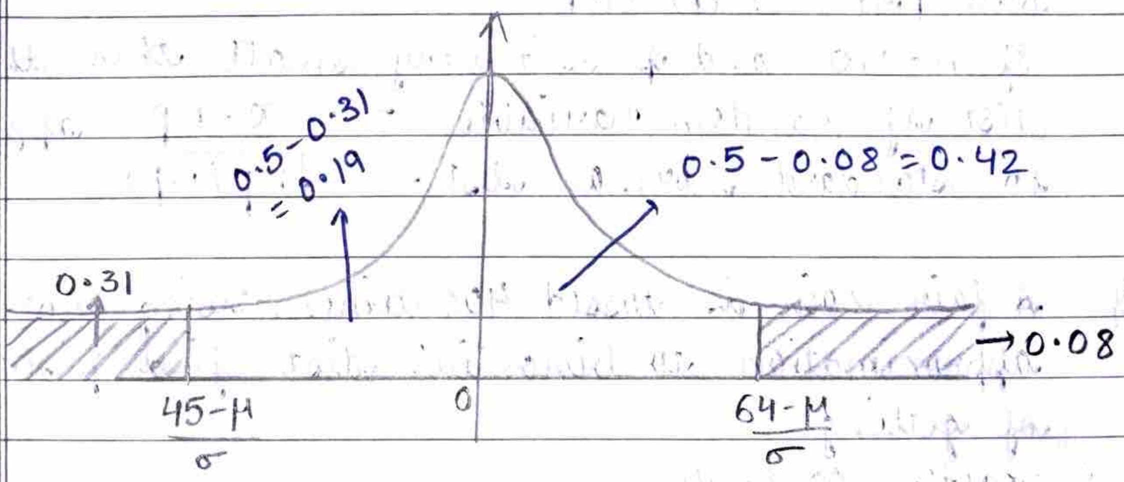
Q In a normal distribution 31% of items are under ₹45 and 8% are over ₹64. Find σ, μ. x is random variable showing cost of item with mean μ and sd σ

$P(x < 45) = 0.31$ and $P(x > 64) = 0.08$

Let

$z_1 = \dfrac{45-\mu}{\sigma}$ & $z_2 = \dfrac{64-\mu}{\sigma}$

$P\left(z < \dfrac{45-\mu}{\sigma}\right) = 0.31$ & $P\left(z > \dfrac{64-\mu}{\sigma}\right) = 0.08$



$$\Rightarrow \int_0^{\frac{-45+\mu}{\sigma}} \frac{1}{\sqrt{2\pi}} e^{-z^2/2}\, dz = 0.19$$

and

$$\int_0^{\frac{64-\mu}{\sigma}} \frac{1}{\sqrt{2\pi}} e^{-z^2/2}\, dz = 0.42$$

we can't directly solve these integrals
so, we use z table
using z table;

$$\frac{-45+\mu}{\sigma} = 0.496$$

$$45 = \mu - 0.496\,\sigma$$

$$\frac{64-\mu}{\sigma} = 1.405$$

$$\Rightarrow \mu + 1.405\,\sigma = 64$$

solve, $\sigma \approx 10$
$\mu \approx 50$

→ Bionomial approximation to normal distribution
Let random variable x follows binomial distribut with PMF $^nC_i\ (p)^i\ (1-p)^{n-i}$
If $n \to \infty$ and p isn't very small then the dist of random variable $z = \frac{x-np}{\sqrt{np(1-p)}}$ approach to standard normal dist.

Q A fair coin is tossed 400 times, using normal approximation to binomial dist find probablity of getting
1) exactly 200 heads
11) between 190 & 200 heads both inclusive
Area b/t $z=0$ & $z=0.05$ is 0.0199 & b/t $z=0$ & $z=1.05$ is 0.3531

$n = 400 \qquad p = 1/2$

$$X \sim B\left(400, \frac{1}{2}\right)$$

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$$z = \frac{x - np}{\sqrt{np(1-p)}}$$

$$np = 400 \times \frac{1}{2} = 200$$

$$z = \frac{x - 200}{\sqrt{200 \times \frac{1}{2}}} = \frac{x-200}{10}$$

1) For 200 heads.

$P(X = 200) \approx P(199.5 < X < 200.5)$

$= P(-0.05 < z < 0.05)$

$= 2 \times P(0 < z < 0.05)$

$= 2 \times 0.0199$

$= 0.0398$

ii) $P(190 \leq X \leq 210)$

$\approx P(189.5 \leq X \leq 210.5)$

$= P(-1.05 \leq Z \leq 1.05)$

$= 2 \times P(0 \leq Z \leq 1.05)$

$= 2 \times 0.3531$

$= 0.7062$

→ Exponential Distribution

A random variable X is said to be having a two variable parameter exp distribution if its PDF is given by

$$f(x) = \begin{cases} \frac{1}{b} e^{-\frac{(x-a)}{b}} & x \geq a \\ 0 & \text{elsewhere} \end{cases}$$

expo. PDF of X

$a > 0$ $b > 0$ are 2 parameters

→ One parameter exponential distribution

$$f(x) = \begin{cases} \lambda e^{-\lambda x} & x \geq 0 \\ 0 & \text{elsewhere} \end{cases}$$

$\lambda > 0$ is the only parameter

- $f(x) \geq 0 \quad \forall x$
- $\int_{-\infty}^{\infty} f(x) = 1$

→ Theorem

If random variable X has exponential dist. with parameter λ then mean $= 1/\lambda = \sigma$

Proof

$$E(x) = \int_0^{\infty} x f(x)\, dx$$

$$= \int_0^{\infty} x \lambda e^{-\lambda x} = \lambda \int_0^{\infty} x e^{-\lambda x} dx$$

$$\lambda x = t$$
$$\lambda dx = dt$$

$$= \int_0^{\infty} \frac{t}{\lambda} e^{-t} dt = \frac{1}{\lambda} \int_0^{\infty} t^{2-1} e^{-t} dt$$

$$\int_0^{\infty} t^{n-1} e^{-t} dt = \Gamma n \rightarrow \text{Gamma}(n)$$

$$\Gamma n = (n-1)!$$

$$= \frac{1}{\lambda} \Gamma 2 = \frac{1}{\lambda} \times 1! = \frac{1}{\lambda}$$

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Q Suppose during rainy season, on a tropical island the length of shower has an exponential distribution with avg. length of 1/2 mins. Whats the probablity that shower lasts more than 3 mins? If a shower has already lasted for 2 mins, probablity that it will last atleast one more minute is?

Let x is random variable depicting the length of shower

$$X \sim \text{exponential}(\lambda)$$

and $\bar{x} = \frac{1}{\lambda} = \frac{1}{2} \Rightarrow \lambda = 2$

$$\Rightarrow f(x) = \begin{cases} 2e^{-2x}, & x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

1) $P(X > 3)$

$$\int_3^\infty f(x)\,dx = \int_3^\infty 2e^{-2x}\,dx$$

$$2x = t$$

$$2\,dx = dt$$

$$\int_6^\infty e^{-t}\,dt = -e^{-\infty} + e^{-6} = e^{-6} - 0$$

$$= e^{-6}$$

ii) $P(X = 2) = \int_0^2 2e^{-2x} = -e^{-4} + e^0 = 1 - e^{-4}$

$P(X \geq 2) = \int_2^\infty 2e^{-2x} = e^{-4}$

Req Probab $= P\left(\frac{x \geq 3}{x \geq 2}\right) = \frac{e^{-6}}{e^{-4}} = e^{-2}$

conditional probab

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Q Let length of a phone call is exponentially dist with mean 10. If someone arrives immediately ahead of you at a telephone booth. Find the probab you have to wait for

1) more than 10 min

11) between 10 & 20

$X \sim \exp(\lambda)$

$\frac{1}{\lambda} = 10, \quad \lambda = 0.1$

$$f(x) = \begin{cases} 0.1\, e^{-0.1x} & x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

1) $P(X > 10) = \int_{10}^{\infty} 0.1\, e^{-0.1x}$

$= -e^{-\infty} + e^{-1}$

$= e^{-1}$

11) $P(10 \leq X \leq 20) = \int_{10}^{20} 0.1\, e^{-0.1x}$

$= e^{-1} - e^{-2}$

→ Gamma Distribution

A random variable X is said to have a gamma distribution with 2 parameters l, λ $l > 0$ $\lambda > 0$ if it's PDF is

$$f(x) = \begin{cases} \dfrac{\lambda e^{-\lambda x} (\lambda x)^{l-1}}{\Gamma l} & x \geqslant 0 \\ 0 & \text{otherwise} \end{cases}$$

If $l = 1$ gama dist = expo dist

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→ Theorem

If $X \sim$ Gamma (λ, ℓ)

then $\mu = \frac{\ell}{\lambda}$ and $\sigma = \frac{\sqrt{\ell}}{\lambda}$

Q If a random variable has gamma dist with parameter $\ell = 2$ & $\lambda = 1/3$ find μ & σ. Hence find probab it takes value less than 5

$$\mu = \frac{\ell}{\lambda} = 6 \text{ and } \sigma = 3\sqrt{2}$$

$$P(X<5) = \int_0^5 \frac{1}{3} \frac{e^{-x/3} (x/3)^1}{\Gamma 2}$$

$$= \frac{1}{9}\int_0^5 x e^{-x/3}$$

$$= \frac{1}{9}\left[\frac{xe^{-x/3}}{-1/3}\Big|_0^5 - \int_0^5 \frac{e^{-x/3}}{-1/3}\right]$$

$$= \frac{1}{9}\left[\frac{5e^{-5/3}}{-1/3} + 3\frac{e^{-x/3}}{-1/3}\Big|_0^5\right]$$

$$= \frac{1}{9}\left[-15e^{-5/3} - 9\left[e^{-5/3} - 1\right]\right]$$

$$= \frac{1}{9}\left[-24 e^{-5/3} + 9\right]$$

$$= 1 - \frac{8}{3}e^{-5/3}$$

$$= 1 - \frac{8}{3}e^{-5/3}$$

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Q In a town, daily consumption of electric power (in million KW) is a random variable having gamma distribution with mean 6 and $\sigma = 2\sqrt{3}$. If power plant has daily capacity of 12 million KW, whats probab that power supply will be inadequate on any given day? For how many days the town would be riled under load shedding in a year

$\frac{l}{\lambda} = 6$ $\lambda = \frac{l}{6}$ $\frac{6\sqrt{l}}{l} = 2\sqrt{3}$

$\lambda = 1/2$ $l = \frac{9}{3} = 3$

$$P(x) = \begin{cases} \frac{1}{2} \frac{e^{-x/2} (x/2)^2}{2} & x \geq 0 \\ 0 \end{cases}$$

$$P(x) = \begin{cases} \frac{1}{16} x e^{-x/2} & x \geq 0 \\ 0 \end{cases}$$

1) $P(X \geq 12) = \int_{12}^{\infty} \frac{1}{16} x e^{-x/2}$

$= \frac{1}{16} \left[-2xe^{-x/2} + \int 2e^{-x/2} \right]_{12}^{\infty}$

$= \frac{1}{16} \left[-2xe^{-x/2} - 4e^{-x/2} \right]_{12}^{\infty}$

$= \frac{-1}{8} \left[(x+1) e^{-x/2} \right]_{12}^{\infty}$

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$= -\frac{1}{8}\left[e^{-\infty} - 13e^{-6}\right]$

$= \frac{13}{8} e^{-6}$

ii) $\frac{13}{8} \times 365\, e^{-6}$

→ Joint Distribution (Bivariate Probablity)

Distribution for a relation between 2 continuous random variables

$f(x, y)$ is bivariate probablity density function if joint dist $F(x, Y)$ is given by

$$F(x, y) = \int_{v=-\infty}^{y} \int_{u=-\infty}^{x} f(u, v)\, du\, dv \quad \forall\ x, y$$

- $f(x, y) \geq 0$ and $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y)\, dx\, dy = 1$

→ Properties of Bivariate Dist

- $P(X = a, Y = b) = 0$ (a & b → constant)
- ★. $P(a < X \leq b,\ c < Y \leq d) = \int_{c}^{d} \int_{a}^{b} f(x, y)\, dx\, dy$
- $\frac{\partial^2 F}{\partial x \partial y} = f(x, y) \quad \forall\ x, y$
- $P(x < X \leq x + dx,\ y < Y \leq y + dy) = dF = f(x, y)$

→ Marginal Distribution

Let (X, Y) is cont. bivariate variable having dist. function $F(x, Y)$ & probablity density $f(x, Y)$. Then marginal dist func of X & Y are

$$F_x(x) = F(x, \infty) = \int_{v=-\infty}^{\infty} \int_{u=-\infty}^{x} f(u, v)\, du\, dv$$

X ka marginal

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and $F_Y(y) = \int_{v=-\infty}^{y} \int_{u=-\infty}^{\infty} f(u,v)\, du\, dv$

The marginal density functions :-

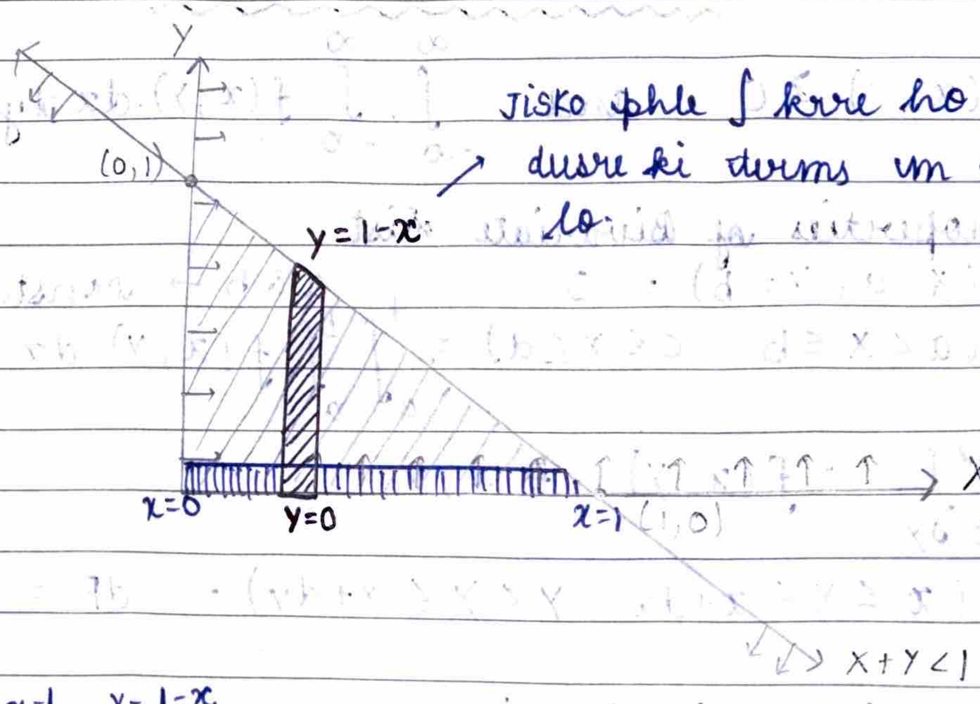
$f_X(x) = \int_{-\infty}^{\infty} f(x,y)\, dy \rightarrow$ x ka nikalte hue y ki limits ayegi & vice versa

$f_Y(y) = \int_{-\infty}^{\infty} f(x,y)\, dx$

$-\infty$ aur ∞ ki jgh x aur y ki min aur max value ayegi

Q Consider a function $f(x,y)$ which is defined as $f(x,y) \begin{cases} K(1-x-y) & \text{for } x>0,\ y>0,\ x+y<1 \\ 0 & \text{elsewhere} \end{cases}$

1. Find K, marginal density functions of X & Y



Jisko phle $\int$ krre ho usko dusre ki terms mein nikal lo.

$\int_{x=0}^{x=1} \int_{y=0}^{y=1-x} K(1-x-y)\, dy\, dx = 1$

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$$K \int_{x=0}^{x=1} \left(y - xy - \frac{y^2}{2} \right)_{y=0}^{1-x} dx = 1$$

$$\Rightarrow K \int_0^1 \left[1-x - x(1-x) - \frac{(1-x)^2}{2} \right] dx = 1$$

$$\Rightarrow K \int_0^1 \left(1 - x - x + x^2 - \frac{1}{2} - \frac{x^2}{2} + \frac{2x}{2} \right) dx = 1$$

$$\Rightarrow K \int_0^1 \left(\frac{1}{2} - x + \frac{x^2}{2} \right) dx = 1$$

$$\Rightarrow K \left[\frac{x}{2} - \frac{x^2}{2} + \frac{x^3}{6} \right]_0^1 = 1$$

$$\Rightarrow K \left[\frac{1}{2} - \frac{1}{2} + \frac{1}{6} \right] = 1$$

$$\Rightarrow K = 6$$

ii) $f_x(x) = \int_0^{1-x} 6(1-x-y)\, dy$

$$f_x(x) = 6 \left[y - xy - \frac{y^2}{2} \right]_0^{1-x}$$

$$= 6 \left(\frac{1}{2} - x + \frac{x^2}{2} \right) = 3(1 - 2x + x^2)$$

$$f_y(y) = \int_0^{1-y} 6(1-x-y)\, dx$$

$$= 6 \left(\frac{1}{2} - y + \frac{y^2}{2} \right) = 3(1 - 2y + y^2)$$

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→ conditional density & conditional distribution

If continuous bivariate (X, Y) has joint PDF $f(x, y)$ then

- conditional probablity function of X given $Y = Y_0$ is given by

$$f_{\frac{X}{Y}}(x / y_0) = \frac{f(x, y_0)}{f_Y(y_0)}$$

||ly,

$$f_{\frac{Y}{X}}(y / x_0) = \frac{f(x_0, y)}{f_X(x_0)}$$

f_Y & f_X are marginal dist functions

If $\underbrace{f(x, y) = f_x(x) \times f_y(y)}$

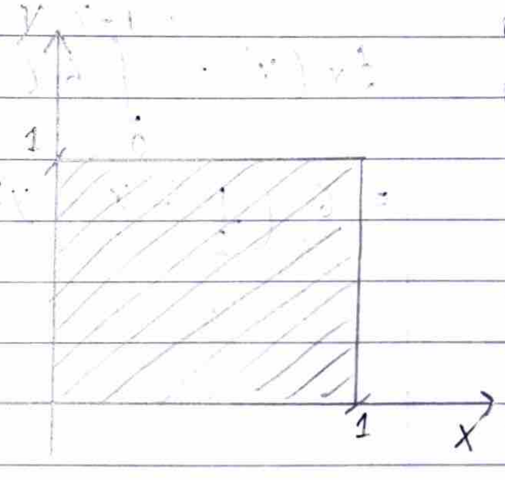
↳ then continous random variables X & Y are said to be independent

PYQ

*Q Let random variables X and Y have joint PDF

$$f(x, y) = \begin{cases} x + y & 0 < x < 1 \quad 0 < y < 1 \\ 0 & \text{elsewhere} \end{cases}$$

Find co-relation coefficient of X, Y



$$f_x(x) = \int_0^1 (x + y)\, dy$$

$$= xy + \frac{y^2}{2} \Big|_0^1$$

$$= x + \frac{1}{2}$$

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||ly $f_Y(y) = y + \frac{1}{2}$

☆ To find expectation of any func $g(x, y)$

$$E[g(x,y)] = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} g(x,y) \cdot f(x,y)\, dx\, dy$$

$$E[g(x)] = \int_{-\infty}^{\infty} g(x) \cdot f_x(x)\, dx$$

Also $cov(x, y) = \frac{\Sigma x_i y_i}{N} - \bar{x}\bar{y}$

when X & Y → discrete

$\exists\ cov(x, y) = E(xy) - E(x)E(y)$

$$E(xy) = \int_{x=0}^{x=1}\int_{y=0}^{y=1} xy\,(x+y)\, dy\, dx$$

$$= \int_{x=0}^{x=1}\int_{y=0}^{y=1} (x^2 y + xy^2)\, dy\, dx$$

$$= \int_{x=0}^{x=1} \left[\frac{x^2y^2}{2} + \frac{xy^3}{3}\right]_0^1 dx = \int_{x=0}^{x=1} \frac{x^2}{2} + \frac{x}{3}\, dx$$

$$= \int_{x=0}^{x=1} \left(\frac{x^2}{2} + \frac{x}{3}\right) dx$$

$$= \frac{x^3}{6} + \frac{x^2}{6}\Big|_0^1$$

$$= \frac{1}{6} + \frac{1}{6}$$

$$= \frac{1}{3}$$

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144
066

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$E(x) = \int_0^1 x\, f_x(x)\, dx$

$= \int_0^1 x\left(x+\frac{1}{2}\right) dx = \int_0^1 x^2 + \frac{x}{2}\; dx$

$= \frac{x^3}{3} + \frac{x^2}{4}\Big|_0^1$

$= \frac{7}{12}$

$E(y) = \int_0^1 y\left(y+\frac{1}{2}\right) dy = \frac{7}{12}$

$\Rightarrow \text{cov}(x, y) = \frac{1}{3} - \frac{7}{12}\times\frac{7}{12} = \frac{1}{3} - \frac{49}{144}$

$= \frac{144 - 147}{432} = \frac{-3}{432} = -\frac{1}{144}$

Now, $var(x) = E(x^2) - (E(x))^2$

$E(x^2) = \int_0^1 x^2\left(x+\frac{1}{2}\right) dx = \int_0^1 \left(x^3 + \frac{x^2}{2}\right) dx$

$= \frac{x^4}{4} + \frac{x^3}{6}\Big|_0^1 = \frac{1}{4} + \frac{1}{6} = \frac{5}{12}$

$E(y^2) = \int_0^1 y^2\left(y+\frac{1}{2}\right) dx = \frac{5}{12}$

$var(x) = \frac{5}{12} - \frac{49}{144} = \frac{60-49}{144} = \frac{11}{144}$

$\Rightarrow \sigma_x = \sqrt{11}/12$

$|||^{ly}$ $\sigma_y = \sqrt{11}/12$

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$$r_{xy} = \frac{cov(x,y)}{\sigma_x \sigma_y} = \frac{-1}{144} \times \frac{12 \times 12}{\sqrt{11} \times \sqrt{11}}$$

$$= \frac{-1}{11}$$

$$= -0.0909$$

Q $f(x,y) = \begin{cases} c(x+y) & x>0 \quad y>0 \quad x+y<2 \\ 0 & \text{elsewhere} \end{cases}$

$c = ?$ $\quad P(x<1, y>\frac{1}{2})$

$$\int_{x=0}^{2} \int_{y=0}^{y=2-x} c(x+y)\, dy\, dx = 1$$

$$c \int_{x=0}^{2} \left[xy + \frac{y^2}{2}\right]_0^{2-x} dx = c \int_0^2 x(2-x) + \frac{(2-x)^2}{2}\, dx$$

$$= c \int_0^2 (2-x)\,[x + 1 - x/2]\, dx = c \int_0^2 (2-x)(1 + x/2)\, dx$$

$$= c \int_0^2 2 + \cancel{x} - \cancel{x} - \frac{x^2}{2}\, dx = c\left[2x - \frac{x^3}{6}\right]_0^2 = 1$$

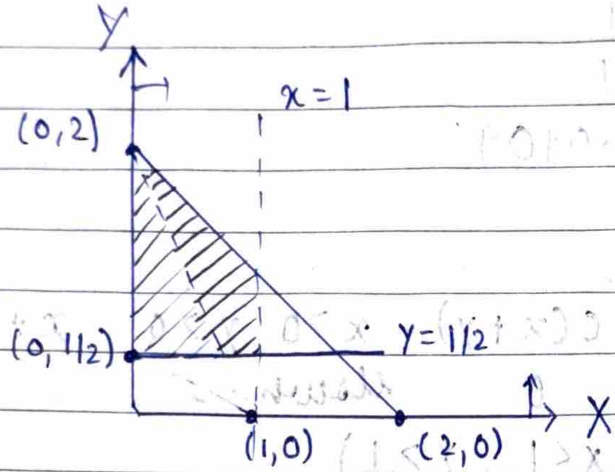
$$= c\left[4 - \frac{8}{6}\right] = 1 \quad \Rightarrow \quad c(8/3) = 1$$

$$\Rightarrow \quad \cancel{c = \frac{16}{11}} \qquad c = \frac{3}{8}$$

$$f(x,y) = \begin{cases} \frac{6}{16}(x+y) & x>0 \quad y>0 \quad x+y<2 \\ 0 & \text{elsewhere} \end{cases}$$

Now

$P(X<1, Y>1/2)$



$P[(0<X<1), (\frac{1}{2}<Y<2-X)]$

$$\int_{x=0}^{x=1}\int_{\frac{1}{2}}^{2-x} \frac{3}{8}(x+y)\,dy\,dx$$

$$\frac{3}{8}\int_0^1 xy+\frac{y^2}{2}\Big|_{\frac{1}{2}}^{2-x} dx = \frac{3}{8}\int_0^1\left[x(2-x)+\frac{(2-x)^2}{2}-\frac{x}{2}-\frac{1}{8}\right]dx$$

$$=\frac{3}{8}\int_0^1 (2-x)\left[\frac{x}{2}+1\right]-\frac{x}{2}-\frac{1}{8}\,dx = \frac{3}{8}\int_0^1 x+2-\frac{x^2}{2}-x-\frac{x}{2}-\frac{1}{8}\,dx$$

$$=\frac{3}{8}\left[2x-\frac{x^3}{6}-\frac{x^2}{4}-\frac{x}{8}\right]_0^1 = \frac{3}{8}\left[2-\frac{1}{6}-\frac{1}{4}-\frac{1}{8}\right]$$

$$=\left(\frac{11}{6}-\frac{3}{8}\right)\frac{3}{8} = \left(\frac{44-9}{24}\right)\frac{3}{8} = \frac{35}{\cancel{24}\,8}\times\frac{\cancel{8}}{8}$$

$$=\frac{35}{64}$$

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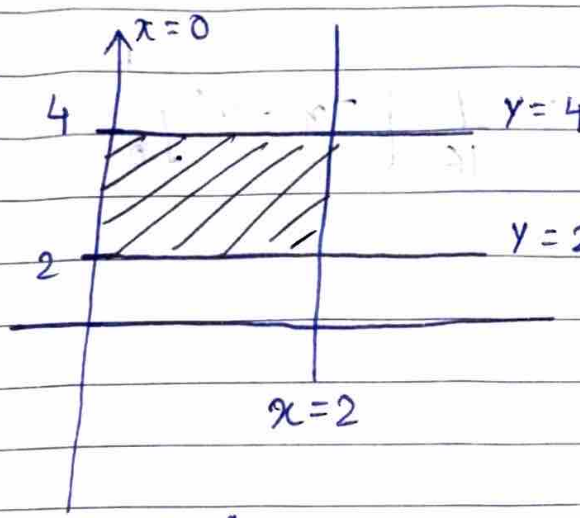
Q $f(x,y) = \begin{cases} \frac{1}{8}(6-x-y) & 0<x<2 \quad 2<y<4 \\ 0 & \end{cases}$

$P(x<1,\ y<3)$, $P(X+Y<3)$, marginal dist

$P\left(\frac{X<1}{Y=3}\right)$ and $P\left(\frac{x<1}{y<3}\right)$



$$f_X(x) = \int_2^4 \frac{1}{8}(6-x-y)\,dy$$

$$= \frac{1}{8}\left[(6-x)y - \frac{y^2}{2}\right]_2^4 = \frac{1}{8}\left[12-2x-6\right] = \frac{1}{8}(6-2x)$$

$$= \frac{1}{4}(3-x) \qquad 0<x<2$$

$$f_Y(y) = \int_0^2 \frac{1}{8}(6-x-y)\,dx$$

$$= \frac{1}{8}\left[(6-y)x - \frac{x^2}{2}\right]_0^2 = \frac{1}{8}\left[12-2y-2\right]$$

$$= \frac{1}{4}\left[5-y\right] \qquad 2<y<4$$

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$\rightarrow P[(X<1), Y<3]$

$= P[0<X<1,\ 2<Y<3]$

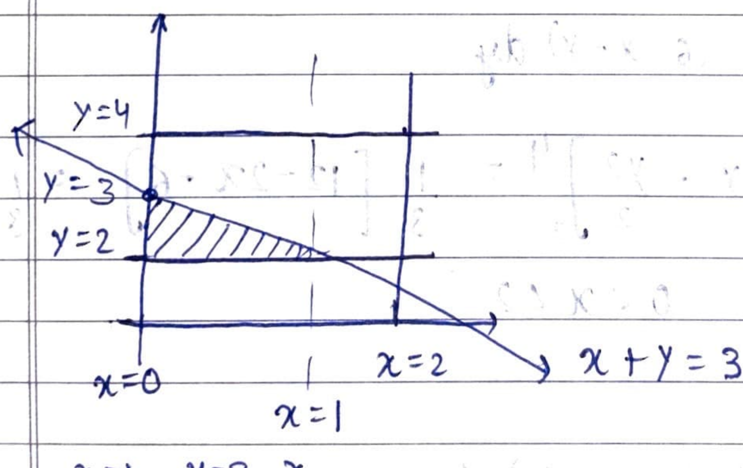
$= \int_{x=0}^{x=1} \int_{y=2}^{y=3} \frac{1}{8}(6-x-y)\, dy\, dx$

$= \frac{1}{8}\int_0^1 6y - xy - \frac{y^2}{2}\Big|_2^3 dx = \frac{1}{8}\int_0^1 \left(6 - x - \frac{5}{2}\right) dx$

$= \frac{1}{16}\int_0^1 7 - 2x\, dx = \frac{1}{16}\left[7x - x^2\right]_0^1$

$= \frac{1}{16}[7-1] = \frac{3}{8}$

$\rightarrow P(X+Y<3)$



$\frac{1}{8}\int_{x=0}^{x=1} \int_{y=2}^{y=3-x} (6-x-y)\, dy\, dx$

$\frac{1}{8}\int_0^1 (6-x)y - \frac{y^2}{2}\Big|_2^{3-x} dx$

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$$\frac{1}{8}\int_0^1 (6-x)(1-x) + 2 - \frac{(3-x)^2}{2}\, dx$$

$$\frac{1}{8}\int_0^1 \left(6 - 7x + x^2 + 2 - \frac{9}{2} - \frac{x^2}{2} + 3x\right) dx$$

$$\frac{1}{8}\int_0^1 \frac{7}{2} - 4x + \frac{x^2}{2}\, dx = \frac{1}{16}\int_0^1 (7 - 8x + x^2)\, dx$$

$$= \frac{1}{16}\left[7x - 4x^2 + \frac{x^3}{3}\right]_0^1 = \frac{1}{16}\left[7 - 4 + \frac{1}{3}\right]$$

$$= \frac{1}{16} \times \frac{10}{3} = \frac{5}{24}$$

$$P\left(\frac{X<1}{Y=3}\right) = \frac{f(x<1, y=3)}{f_Y(y=3)} = \frac{f(0<x<1, y=3)}{f_Y(y=3)}$$

$$= \int_0^1 \frac{(6-x-3)\,1/8}{(5-3)\,1/4}\, dx = \frac{1}{4}\int_0^1 3 - x \quad dx$$

$$= \frac{1}{4}\left[3x - \frac{x^2}{2}\right]_0^1 = \frac{1}{4}\left[3 - \frac{1}{2}\right] = \frac{5}{8}$$

$$\Rightarrow P\left(\frac{0<X<1}{2<Y<3}\right) = \frac{f(0<x<1, 2<y<3)}{f_Y(y)}$$

$$= \frac{\frac{1}{8}\int_{x=0}^{x=1}\int_{y=2}^{y=3} 6 - x - y \; dy\, dx}{\frac{1}{4}\int_2^3 5 - y \; dy} = \frac{\frac{3}{8}}{\frac{\cancel{10}\,5}{\cancel{16}\,8}} = \frac{3}{5}$$

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→ Chebyshev's inequality

It states that for a wide class of probablity distribution, no more than a certain fraction of value can be more than a certain distance from the mean

Acc to chebyshev's inequality

- $P(|x-\mu| < K\sigma) \geq 1-\frac{1}{K^2}$
- $P(|x-\mu| \geq K\sigma) \leq \frac{1}{K^2}$

$\mu \rightarrow$ mean

$K \rightarrow$ constant

Q A random variable X has mean 8 & variance 9, Find

1) $P(-4 < x < 20)$
2) $P(|x-8| \geq 6)$

$\mu = 8$

$\sigma = 3$

1)

$= P(-4 < X < 20)$

$= P(-4-8 < X-8 < 20-8) = P(-12 < x-\mu < 12)$

$= P(|x-\mu| < 12)$

$\Rightarrow K\sigma = 12$

$\Rightarrow K = 4$

$P(|x-\mu| < 12) \geq 1-\frac{1}{K^2}$

$P(|x-\mu| < 12) \geq \frac{15}{16}$

2) $P(|x-8| \geq 6)$

$K\sigma = 6$

$K = 3$

$P(|x-8| \geq 6) \leq \frac{1}{K^2} \leq \frac{1}{9}$

Q Determine the smallest value of K in which Chebyshev's theorem

1. Probablity is atleast 0.95
2. Probablity is atleast 0.99

☆ Chebyshev's theorem states that for any random variable X with mean μ and sd σ, the probablity that x falls within K standard deviations of mean is atleast $1 - \frac{1}{K^2}$

$$P(|X-\mu| < K\sigma) \geq 1 - \frac{1}{K^2}$$

1) For P to be atleast 0.95

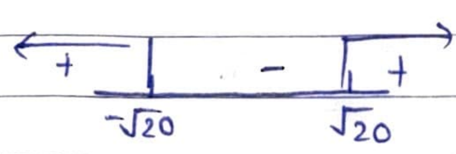
$$1 - \frac{1}{K^2} \geq 0.95$$

$$\frac{1}{K^2} \leq 0.05$$

$$K^2 \geq 20$$

$$K^2 - 20 \geq 0$$

$$(K - \sqrt{20})(K + \sqrt{20}) \geq 0$$



But $1 - \frac{1}{K^2} \leq 1$ $\because$ it represents probablity

$\Rightarrow K^2 \geq 0$

Hence $K \in [\sqrt{20}, \infty)$

smallest value of $K = \sqrt{20} = 4.47$

2) IIly $1 - \frac{1}{K^2} \geq 0.99$

$\Rightarrow K = 10$